

SQL Analysis Results

SECTION 1: Loyal vs Discount Customers

Business Question

Who buys without needing a discount, and can these customers support a promo-sunset strategy?

Analysis 1: Overall Promo Dependency Split

Objective:

Determine how many customers rely on discounts and compare their spending, purchase history, and customer value.

SQL Query:

```
SELECT
    Promo_Dependent,
    COUNT(*) AS customer_count,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*)
        FROM customers), 1) AS pct_of_total,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Frequency_Score), 1) AS avg_frequency_score,
    ROUND(AVG(Customer_Value_Score), 3) AS avg_value_score
FROM customers
GROUP BY Promo_Dependent
ORDER BY Promo_Dependent DESC;
```

Result:

	Promo_Dependent	customer_count	pct_of_total	avg_spend	avg_prev_purchases	avg_frequency_score	avg_value_score
►	1	1677	43.0	59.28	25.7	17.6	0.43
	0	2223	57.0	60.13	25.1	17.4	0.425

Key Findings

- 57% of customers are non-promo-dependent, while 43% rely on discounts.
- Average spend is almost identical between both groups (~\$60), showing discounts do not strongly impact spending.

- Non-promo-dependent customers show slightly higher purchase frequency and value score.

Conclusion

A majority of customers already purchase without discounts, making them ideal for a promo-sunset strategy without risking revenue loss.

Analysis 2: Loyalty vs Promo Dependency

Objective:

Understand how loyalty levels differ between promo-dependent and non-promo-dependent customers.

SQL Query:

```
SELECT
    Loyalty_Label_A,
    Promo_Dependent,
    COUNT(*) AS customer_count,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Review_Rating), 2) AS avg_rating
FROM customers
GROUP BY Loyalty_Label_A, Promo_Dependent
ORDER BY Loyalty_Label_A, Promo_Dependent DESC;
```

Result:

Loyalty_Label_A	Promo_Dependent	customer_count	avg_spend	avg_prev_purchases	avg_rating
Developing	1	499	60.23	33.4	3.71
Developing	0	790	59.50	20.6	3.74
Loyal	1	262	58.02	37.9	3.74
Loyal	0	1074	60.47	34.2	3.77
Transactional	1	916	59.12	18.1	3.76
Transactional	0	359	60.49	7.5	3.75

Key Findings

- Loyal customers show higher purchase frequency and purchase history than Transactional customers.

- Non-promo-dependent Loyal customers have the highest average spend (\$60.47).
- Transactional customers have the weakest engagement and lowest purchase history.

Conclusion

Loyal customers generate strong value without relying on discounts, making them the best target for retention and promo-sunset strategies.

Analysis 3: Promo Dependency by Loyalty Tier

Objective:

Measure discount reliance across each loyalty segment.

SQL Query:

```
SELECT
    Loyalty_Label_A,
    COUNT(*) AS total,
    SUM(Promo_Dependent) AS promo_users,
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS promo_pct
FROM customers
GROUP BY Loyalty_Label_A
ORDER BY promo_pct DESC;
```

Result:

Loyalty_Label_A	total	promo_users	promo_pct
Transactional	1275	916	71.8
Developing	1289	499	38.7
Loyal	1336	262	19.6

Key Findings

- Transactional customers have the highest promo dependency (71.8%).
- Developing customers show moderate promo dependency (38.7%).
- Loyal customers have the lowest promo dependency (19.6%), indicating they purchase without relying on discounts.

Conclusion

Promo dependency decreases as loyalty increases. Loyal customers are the best candidates for a promo-sunset strategy, while Transactional customers remain highly discount-driven.

Analysis 4: Promo Value Segment Deep Dive

Objective:

Identify which customer groups provide the highest value relative to promotional usage.

SQL Query:

```
SELECT
    Promo_Value_Segment,
    COUNT(*) AS customer_count,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Customer_Value_Score), 3) AS avg_cvs,
    ROUND(AVG(Review_Rating), 2) AS avg_rating
FROM customers
GROUP BY Promo_Value_Segment
ORDER BY avg_cvs DESC;
```

Result:

Promo_Value_Segment	customer_count	avg_spend	avg_prev_purchases	avg_cvs	avg_rating
High_Value_Promo_Dependent	725	66.23	34.2	0.606	3.74
High_Value_Organic	930	67.14	33.8	0.601	3.77
Low_Value_Organic	1293	55.09	18.8	0.299	3.75
Low_Value_Promo_Dependent	952	53.98	19.3	0.296	3.75

Key Findings

- High-Value segments have the highest customer value scores (~0.60) and average spend above \$66.
- High-Value Organic customers slightly outperform High-Value Promo-Dependent customers in spend and rating.
- Low-Value segments show significantly lower spend (~\$54–55) and customer value scores (~0.30).

Conclusion

High-Value Organic customers represent the strongest customer group because they generate high value without relying heavily on promotions. They should be prioritized for retention and used as the benchmark for future customer acquisition.

SECTION 2: High Value Customer Analysis

Business Question

What separates high-value customers from low-value customers?

Analysis 1: High Value vs Low Value Full Profile Comparison

Objective:

Compare spending, purchase behavior, subscriptions, and engagement between high-value and low-value customers.

SQL Query:

```
SELECT
    High_Value_Flag,
    COUNT(*) AS customer_count,
    ROUND(AVG(Age), 1) AS avg_age,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Frequency_Score), 1) AS avg_frequency,
    ROUND(AVG(Review_Rating), 2) AS avg_rating,
    ROUND(AVG(Subscription_Status), 2) AS subscription_rate,
    ROUND(AVG(Promo_Dependent), 2) AS promo_dependency_rate,
    ROUND(AVG(Premium_Shipping_Flag), 2) AS premium_shipping_rate
FROM customers
GROUP BY High_Value_Flag
ORDER BY High_Value_Flag DESC;
```

Result:

High_Value_Flag	customer_count	avg_age	avg_spend	avg_prev_purchases	avg_frequency	avg_rating	subscription_rate	promo_dependency_rate	premium_shipping_rate
1	1655	44.4	66.74	34.0	28.7	3.75	0.29	0.44	0.51
0	2245	43.8	54.62	19.0	9.2	3.75	0.26	0.42	0.48

Key Findings

- High-value customers spend more (**\$66.74 vs \$54.62**).
- They have a stronger purchase history (**34 vs 19 previous purchases**).

- They buy more frequently (**28.7 vs 9.2 purchases/year**).
- Subscription and premium shipping usage are slightly higher among high-value customers.

Conclusion

High-value customers are primarily differentiated by repeat purchases and purchase frequency. Retention strategies should focus on keeping these customers engaged, as they generate significantly higher revenue for the business.

Analysis 2: Value Tier Deep Dive

Objective:

Analyze differences across customer value tiers and identify the characteristics of premium customers.

SQL Query:

```
SELECT
    Value_Tier,
    COUNT(*) AS customer_count,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Frequency_Score), 1) AS avg_frequency,
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS promo_pct,
    ROUND(AVG(Review_Rating), 2) AS avg_rating,
    ROUND(AVG(Subscription_Status) * 100, 1) AS subscription_pct
FROM customers
GROUP BY Value_Tier
ORDER BY avg_spend DESC;
```

Result:

Value_Tier	customer_count	avg_spend	avg_prev_purchases	avg_frequency	promo_pct	avg_rating	subscription_pct
Platinum	503	69.78	37.3	43.6	44.5	3.78	28.6
Gold	1152	65.42	32.5	22.2	43.5	3.74	28.8
Silver	1560	57.39	23.1	11.2	42.1	3.75	25.8
Bronze	685	48.30	9.7	4.7	43.1	3.74	25.5

Key Findings

- Customer value increases consistently from **Bronze** → **Silver** → **Gold** → **Platinum**.
- **Platinum customers** have the highest spend (**\$69.78**), purchase history (**37.3 purchases**), and purchase frequency (**43.6/year**).

- Promo dependency remains relatively similar across tiers (**42–45%**), suggesting high-value customers are not significantly more discount-dependent.
- Subscription rates are slightly higher in Gold and Platinum tiers.

Conclusion

Platinum and Gold customers represent the most valuable customer segments due to their higher spending and stronger purchasing behavior. Retention efforts should prioritize these tiers, while Bronze and Silver customers can be targeted for value-growth campaigns.

Analysis 3: Subscription Status Impact on Value

Objective:

Evaluate whether subscription status influences customer value and repeat purchase behavior.

SQL Query:

```
SELECT
    Subscription_Status,
    Value_Tier,
    COUNT(*) AS customer_count,
    ROUND(AVG(Customer_Value_Score), 3) AS avg_cvs,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases
FROM customers
GROUP BY Subscription_Status, Value_Tier
ORDER BY Subscription_Status DESC, avg_cvs DESC;
```

Result:

Subscription_Status	Value_Tier	customer_count	avg_cvs	avg_prev_purchases
1	Platinum	144	0.76	37.1
1	Gold	332	0.538	32.6
1	Silver	402	0.352	23.6
1	Bronze	175	0.172	10.5
0	Platinum	359	0.754	37.4
0	Gold	820	0.536	32.5
0	Silver	1158	0.354	22.9
0	Bronze	510	0.17	9.5

Key Findings

- Customer Value Score (CVS) and purchase history are almost identical between subscribers and non-subscribers within each value tier.
- **Platinum customers** have the highest value score (~0.76) and strongest purchase history (~37 purchases) regardless of subscription status.
- The biggest driver of customer value appears to be **Value Tier**, not subscription status.
- Subscription membership alone does not significantly increase customer value or repeat purchases.

Conclusion

Subscription status has limited impact on customer value compared to underlying customer behavior. Retention and growth strategies should focus on high-value customer segments (especially Platinum and Gold) rather than relying solely on subscription programs.

SECTION 3: Revenue Concentration

Business Question

How concentrated is customer value and revenue across different customer groups?

Analysis 1: Revenue Share by Value Tier

Objective:

Determine which value tiers contribute the largest share of total revenue and whether revenue is concentrated among premium customers.

SQL Query:

```
SELECT
    Value_Tier,
    COUNT(*)                                AS customer_count,
    ROUND(SUM(Purchase_Amount), 0)          AS total_spend,
    ROUND(AVG(Purchase_Amount), 2)          AS avg_spend,
    ROUND(SUM(Purchase_Amount) * 100.0 /
        (SELECT SUM(Purchase_Amount)
         FROM customers), 1)                AS revenue_share_pct,
    ROUND(COUNT(*) * 100.0 /
        (SELECT COUNT(*) FROM customers), 1) AS customer_share_pct
FROM customers
GROUP BY Value_Tier
ORDER BY total_spend DESC;
```


Result:

Value_Tier	customer_count	total_spend	avg_spend	revenue_share_pct	customer_share_pct
Silver	1560	89536	57.39	38.4	40.0
Gold	1152	75364	65.42	32.3	29.5
Platinum	503	35097	69.78	15.1	12.9
Bronze	685	33084	48.30	14.2	17.6

Key Findings

- Silver customers contribute the most revenue (**38.4%**) due to their large customer base.
- Platinum customers have the highest average spend (**\$69.78**).
- Gold + Platinum customers generate nearly **45% of total revenue**.

Conclusion

Gold and Platinum customers are the most valuable segments, while Silver customers drive revenue through volume. Retention efforts should focus on Gold and Platinum tiers.

Analysis 2: Revenue Share by Loyalty Label

Objective:

Measure how much revenue is generated by each loyalty segment.

SQL Query:

```
SELECT
    Loyalty_Label_A,
    COUNT(*) AS customers,
    ROUND(SUM(Purchase_Amount), 0) AS total_revenue,
    ROUND(SUM(Purchase_Amount) * 100.0 /
        (SELECT SUM(Purchase_Amount)
         FROM customers), 1) AS revenue_pct,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend
FROM customers
GROUP BY Loyalty_Label_A
ORDER BY total_revenue DESC;
```

Result:

Loyalty_Label_A	customers	total_revenue	revenue_pct	avg_spend
Loyal	1336	80150	34.4	59.99
Developing	1289	77060	33.1	59.78
Transactional	1275	75871	32.6	59.51

Key Findings

- Revenue is evenly distributed across loyalty segments (**Loyal: 34.4%, Developing: 33.1%, Transactional: 32.6%**).
- Average spend is nearly identical across all groups (~\$60).
- No single loyalty segment dominates revenue generation.

Conclusion

All loyalty segments contribute similarly to revenue. The business should focus on moving customers from Transactional and Developing into Loyal segments rather than relying on one segment alone.

Analysis 3: Estimated Promo Cost Impact

Objective:

Estimate the revenue impact of discount-driven purchases and potential savings from reducing promotional dependency.

SQL Query:

```
SELECT
    Promo_Dependent,
    COUNT(*)                                AS customers,
    ROUND(SUM(Purchase_Amount), 0)         AS gross_revenue,
    ROUND(SUM(Purchase_Amount) * 0.15, 0)  AS estimated_discount_cost,
    ROUND(SUM(Purchase_Amount) * 0.85, 0)  AS estimated_net_revenue
FROM customers
GROUP BY Promo_Dependent;
```

Result:

Promo_Dependent	customers	gross_revenue	estimated_discount_cost	estimated_net_revenue
1	1677	99411	14912	84499
0	2223	133670	20051	113620

Key Findings

- Promo-dependent customers generate **\$99,411** in revenue and incur an estimated **\$14,912** discount cost.
- Non-promo-dependent customers generate higher revenue (**\$133,670**) without requiring discounts.
- Reducing promo dependency could improve profitability while maintaining strong revenue.

Conclusion

Non-promo-dependent customers generate more revenue with lower marketing cost. The business should gradually reduce discount reliance and focus on retaining organic customers.

SECTION 4: Geographic Analysis

Business Question

Which states generate the strongest organic demand and where should marketing investments be prioritized?

Analysis 1: State Performance Overview

Objective:

Compare states based on spending, customer value, purchase history, ratings, and promo dependency.

SQL Query:

```
SELECT
    Location
    COUNT(*)
    ROUND(AVG(Purchase_Amount), 2)
    ROUND(AVG(Previous_Purchases), 1)
    ROUND(AVG(Customer_Value_Score), 3)
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1)
    ROUND(AVG(Review_Rating), 2)
    AS state,
    AS customer_count,
    AS avg_spend,
    AS avg_prev_purchases,
    AS avg_cvs,
    AS promo_dependency_pct,
    AS avg_rating
FROM customers
GROUP BY Location
ORDER BY avg_cvs DESC
LIMIT 20;
```

Result:

state	customer_count	avg_spend	avg_prev_purchases	avg_cvs	promo_dependency_pct	avg_rating
Hawaii	65	57.72	29.2	0.491	49.2	3.68
Alaska	72	67.60	28.1	0.484	40.3	3.8
Pennsylvania	74	66.57	27.4	0.468	44.6	3.78
Wyoming	71	60.69	28.2	0.461	42.3	3.69
Arizona	65	66.55	28.4	0.46	33.8	3.69
Illinois	92	61.05	26.6	0.459	40.2	3.64
Tennessee	77	61.97	26.0	0.457	36.4	3.78
Iowa	69	60.88	27.6	0.452	52.2	3.83
Alabama	89	59.11	27.4	0.449	40.4	3.78
Indiana	79	58.92	25.8	0.447	57.0	3.81
Georgia	79	58.80	25.1	0.447	40.5	3.62
North Carolina	78	60.79	24.8	0.443	44.9	3.81
Arkansas	79	61.11	27.1	0.441	46.8	3.71
Nevada	87	63.38	26.0	0.441	47.1	3.72
New Mexico	81	61.90	26.0	0.439	44.4	3.73
Oregon	74	57.34	25.5	0.437	51.4	3.75
Mississippi	80	61.04	26.0	0.437	47.5	3.76
Kentucky	79	55.72	26.4	0.435	43.0	3.74
Missouri	81	57.91	28.1	0.435	49.4	3.77
Louisiana	84	57.71	23.4	0.434	41.7	3.78

Key Findings

- Hawaii shows high customer value with slightly above-average promo dependency (**49.2%**). Alaska shows similarly high customer value but below-average promo dependency (**40.3%**), making it a stronger organic-demand state.
- Several states (e.g., Indiana, Iowa) rely heavily on discounts (>50% promo usage).
- Spending is fairly consistent across states.

Conclusion

Geography does not strongly impact spending, but it affects promo behavior. Focus on reducing discount dependence in high-value states.

Analysis 2: Geographic Opportunity Matrix

Objective:

Identify high-spend organic markets versus discount-driven markets.

SQL Query:

```
SELECT
    Location
    COUNT(*)
    ROUND(AVG(Purchase_Amount), 2)
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1)
    ROUND(AVG(Customer_Value_Score), 3)
)
CASE
    WHEN AVG(Purchase_Amount) >= 60
        AND SUM(Promo_Dependent) * 100.0 / COUNT(*) < 50
    THEN 'Priority – High Spend Organic'
    WHEN AVG(Purchase_Amount) >= 60
        AND SUM(Promo_Dependent) * 100.0 / COUNT(*) >= 50
    THEN 'Discount Market – High Spend Promo Driven'
    WHEN AVG(Purchase_Amount) < 60
        AND SUM(Promo_Dependent) * 100.0 / COUNT(*) < 50
    THEN 'Growth Market – Low Spend Organic'
    ELSE 'Low Priority – Low Spend High Promo'
END
AS market_type
FROM customers
GROUP BY Location
HAVING COUNT(*) >= 20
ORDER BY avg_spend DESC;
```

Result:

state	customers	avg_spend	promo_pct	avg_cvs	market_type
Alaska	72	67.6	40.3	0.484	Priority – High Spend Organic
Pennsylvania	74	66.57	44.6	0.468	Priority – High Spend Organic
Arizona	65	66.55	33.8	0.46	Priority – High Spend Organic
West Virginia	81	63.88	49.4	0.421	Priority – High Spend Organic
Nevada	87	63.38	47.1	0.441	Priority – High Spend Organic
Washington	73	63.33	43.8	0.417	Priority – High Spend Organic
North Dakota	83	62.89	45.8	0.402	Priority – High Spend Organic
Virginia	77	62.88	37.7	0.429	Priority – High Spend Organic
Utah	71	62.58	46.5	0.432	Priority – High Spend Organic
Michigan	73	62.1	39.7	0.422	Priority – High Spend Organic
Tennessee	77	61.97	36.4	0.457	Priority – High Spend Organic
New Mexico	81	61.9	44.4	0.439	Priority – High Spend Organic
Rhode Island	63	61.44	39.7	0.425	Priority – High Spend Organic
Texas	77	61.19	36.4	0.398	Priority – High Spend Organic
Arkansas	79	61.11	46.8	0.441	Priority – High Spend Organic
Illinois	92	61.05	40.2	0.459	Priority – High Spend Organic
Mississippi	80	61.04	47.5	0.437	Priority – High Spend Organic
Massachusetts	72	60.89	48.6	0.428	Priority – High Spend Organic
Iowa	69	60.88	52.2	0.452	Discount Market – High Spend Promo Driven
North Carolina	78	60.79	44.9	0.443	Priority – High Spend Organic
Wyoming	71	60.69	42.3	0.461	Priority – High Spend Organic
South Dakota	70	60.51	37.1	0.415	Priority – High Spend Organic
New York	87	60.43	41.4	0.434	Priority – High Spend Organic
Ohio	77	60.38	44.2	0.427	Priority – High Spend Organic
Montana	96	60.25	37.5	0.418	Priority – High Spend Organic
Idaho	93	60.08	40.9	0.404	Priority – High Spend Organic
Nebraska	87	59.45	42.5	0.415	Growth Market – Low Spend Organic
New Hampshire	71	59.42	43.7	0.42	Growth Market – Low Spend Organic
Alabama	89	59.11	40.4	0.449	Growth Market – Low Spend Organic
California	95	59	42.1	0.413	Growth Market – Low Spend Organic
Indiana	79	58.92	57	0.447	Low Priority – Low Spend High Promo
Georgia	79	58.8	40.5	0.447	Growth Market – Low Spend Organic
South Carolina	76	58.41	50	0.4	Low Priority – Low Spend High Promo
Oklahoma	75	58.35	48	0.423	Growth Market – Low Spend Organic
Missouri	81	57.91	49.4	0.435	Growth Market – Low Spend Organic
Hawaii	65	57.72	49.2	0.491	Growth Market – Low Spend Organic
Louisiana	84	57.71	41.7	0.434	Growth Market – Low Spend Organic
Oregon	74	57.34	51.4	0.437	Low Priority – Low Spend High Promo
Vermont	85	57.18	38.8	0.397	Growth Market – Low Spend Organic
Maine	77	56.99	35.1	0.389	Growth Market – Low Spend Organic
New Jersey	67	56.75	40.3	0.393	Growth Market – Low Spend Organic
Minnesota	88	56.56	44.3	0.434	Growth Market – Low Spend Organic
Colorado	75	56.29	41.3	0.397	Growth Market – Low Spend Organic
Wisconsin	75	55.95	48	0.378	Growth Market – Low Spend Organic
Florida	68	55.85	45.6	0.414	Growth Market – Low Spend Organic
Maryland	86	55.76	44.2	0.434	Growth Market – Low Spend Organic
Kentucky	79	55.72	43	0.435	Growth Market – Low Spend Organic
Delaware	86	55.33	45.3	0.397	Growth Market – Low Spend Organic
Kansas	63	54.56	23.8	0.407	Growth Market – Low Spend Organic

Key Findings

- Most states fall into the **"Priority – High Spend Organic"** category, indicating strong spending with relatively low promo dependency.

- Only a few states are classified as **"High Spend Promo Driven"**, suggesting limited reliance on discounts among top-spending markets.
- Several states are categorized as **"Growth Market – Low Spend Organic"**, offering potential for future expansion.

Conclusion

The business has many high-spend organic markets that can support a promo-sunset strategy. Marketing investment should prioritize these states while reducing discount reliance in promo-driven markets.

Analysis 3: Top Revenue States

Objective:

Identify the states contributing the highest total revenue.

SQL Query:

```
SELECT
    Location,
    COUNT(*)                                AS customers,
    ROUND(SUM(Purchase_Amount), 0)          AS total_spend,
    ROUND(AVG(Purchase_Amount), 2)          AS avg_spend,
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS promo_pct
FROM customers
GROUP BY Location
ORDER BY total_spend DESC
LIMIT 10;
```

Result:

Location	customers	total_spend	avg_spend	promo_pct
Montana	96	5784	60.25	37.5
Illinois	92	5617	61.05	40.2
California	95	5605	59.00	42.1
Idaho	93	5587	60.08	40.9
Nevada	87	5514	63.38	47.1
Alabama	89	5261	59.11	40.4
New York	87	5257	60.43	41.4
North Dakota	83	5220	62.89	45.8
West Virginia	81	5174	63.88	49.4
Nebraska	87	5172	59.45	42.5

Key Findings

- Montana, Illinois, and California are the top revenue-generating states.
- Top revenue states show strong average spending (~\$59–64) per customer.
- Promo dependency remains moderate (~38–49%) across these high-revenue markets.

Conclusion

These states are the business's strongest revenue contributors and should be prioritized for retention and targeted marketing initiatives.

SECTION 5: Category Analysis

Business Question

Which product categories acquire new customers and which categories drive long-term retention?

Analysis 1: Category Performance Overview

Objective:

Compare categories based on spending, customer value, purchase history, ratings, and promo dependency.

SQL Query:

```
SELECT
    Category,
    COUNT(*) AS customer_count,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Customer_Value_Score), 3) AS avg_cvs,
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS promo_dependency_pct,
    ROUND(AVG(Review_Rating), 2) AS avg_rating,
    ROUND(AVG(Frequency_Score), 1) AS avg_frequency
FROM customers
GROUP BY Category
ORDER BY avg_prev_purchases DESC;
```

Result:

Category	customer_count	avg_spend	avg_prev_purchases	avg_cvs	promo_dependency_pct	avg_rating	avg_frequency
Accessories	1240	59.84	25.7	0.429	43.8	3.77	17.3
Clothing	1737	60.03	25.2	0.426	42.1	3.72	17.4
Footwear	599	60.26	25.2	0.431	43.2	3.79	17.9
Outerwear	324	57.17	25.0	0.419	44.4	3.75	17.7

Key Findings

- **Footwear** has the highest customer value score (0.431) and purchase frequency (17.9).
- **Accessories and Clothing** have the largest customer bases and similar spending levels (~\$60).
- Promo dependency is similar across all categories (**42–44%**).

Conclusion

All categories perform similarly, but Footwear stands out slightly in customer value and engagement, making it a strong category for retention and upselling efforts.

Analysis 2: Entry Point vs Retention Categories

Objective:

Identify which categories attract new customers and which are associated with repeat purchasing behavior.

SQL Query:

```
SELECT
    Category,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS promo_pct,
    CASE
        WHEN AVG(Previous_Purchases) < 20
        THEN 'Entry Point Category'
        WHEN AVG(Previous_Purchases) < 30
        THEN 'Mid Journey Category'
        ELSE 'Retention Category'
    END AS category_role
FROM customers
GROUP BY Category
ORDER BY avg_prev_purchases ASC;
```


Result:

Category	avg_prev_purchases	avg_spend	promo_pct	category_role
Outerwear	25.0	57.17	44.4	Mid Journey Category
Clothing	25.2	60.03	42.1	Mid Journey Category
Footwear	25.2	60.26	43.2	Mid Journey Category
Accessories	25.7	59.84	43.8	Mid Journey Category

Key Findings

- All categories are classified as Mid Journey Categories, with average previous purchases around 25–26.
- Accessories have the highest purchase history (25.7 purchases), while Outerwear has the lowest (25.0 purchases).
- Spending and promo dependency are very similar across categories.

Conclusion

No category clearly acts as a pure entry-point or retention category. Customer behavior is relatively consistent across product categories, suggesting retention strategies can be applied broadly rather than category-specific.

Analysis 3: Category × Season Analysis

Objective:

Understand how customer behavior changes across categories and seasons.

SQL Query:

```
SELECT
    Category,
    Season,
    COUNT(*) AS customers,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS promo_pct
FROM customers
GROUP BY Category, Season
ORDER BY Category, avg_prev_purchases ASC;
```

Result:

Category	Season	customers	avg_prev_purchases	avg_spend	promo_pct
Accessories	Fall	324	24.7	61.34	42.6
Accessories	Winter	303	25.5	60.37	43.6
Accessories	Spring	301	25.9	56.50	46.5
Accessories	Summer	312	26.8	60.99	42.6
Clothing	Summer	408	24.5	56.56	45.1
Clothing	Spring	454	24.8	61.00	44.9
Clothing	Fall	427	25.5	61.41	35.6
Clothing	Winter	448	26.1	60.88	42.6
Footwear	Spring	163	24.2	58.62	38.7
Footwear	Fall	136	24.6	63.71	49.3
Footwear	Summer	160	25.6	58.71	45.6
Footwear	Winter	140	26.7	60.57	40.0
Outerwear	Summer	75	24.3	57.04	44.0
Outerwear	Fall	88	24.4	59.76	45.5
Outerwear	Winter	80	25.2	57.03	47.5
Outerwear	Spring	81	26.0	54.63	40.7

Key Findings

- Across all categories and seasons, **behavior is highly consistent** with no major seasonal spikes in spend or purchase history.
- Average spend stays stable around **~\$55–\$61** across seasons.
- Promo dependency remains steady (**~40–47%**) regardless of season or category.
- Slight variations exist (e.g., Winter/Spring slightly higher in some categories like Footwear), but not significant.

Conclusion

Seasonality does not strongly influence customer behavior in this dataset. Marketing and retention strategies can be planned more on category and customer value rather than season-based targeting.

Analysis 4: Category × Loyalty Analysis

Objective:

Examine how loyalty levels differ across product categories.

SQL Query:

```
SELECT
    Category,
    Loyalty_Label_A,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases
FROM customers
GROUP BY Category, Loyalty_Label_A
ORDER BY Category, avg_prev_purchases DESC;
```

Result:

Category	Loyalty_Label_A	customers	avg_spend	avg_prev_purchases
Accessories	Loyal	423	59.93	35.5
Accessories	Developing	422	61.08	25.8
Accessories	Transactional	395	58.42	15.2
Clothing	Loyal	594	60.23	34.6
Clothing	Developing	569	59.70	25.5
Clothing	Transactional	574	60.13	15.2
Footwear	Loyal	211	60.20	34.3
Footwear	Developing	191	59.77	25.9
Footwear	Transactional	197	60.79	14.9
Outerwear	Loyal	108	58.53	35.9
Outerwear	Developing	107	55.12	24.4
Outerwear	Transactional	109	57.84	14.6

Key Findings

- Loyal customers consistently have the highest purchase history (~34–36 purchases) across all categories.
- Average spend is very similar across loyalty groups (~\$58–61), regardless of category.
- Clothing and Accessories contain the largest number of loyal customers.

Conclusion

Loyalty is driven more by repeat purchasing behavior than spending amount. Retention efforts should focus on increasing purchase frequency across all categories.

SECTION 6: Demographic Analysis

Business Question

Which demographic groups offer the greatest opportunity for growth and retention?

Analysis 1: Age Group Performance

Objective:

Compare customer value, spending, purchase behavior, subscriptions, and promo dependency across age groups.

SQL Query:

```
SELECT
  Age_Group,
  COUNT(*) AS customers,
  ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
  ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
  ROUND(AVG(Customer_Value_Score), 3) AS avg_cvs,
  ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS promo_pct,
  ROUND(AVG(Subscription_Status) * 100, 1) AS subscription_pct,
  ROUND(AVG(Review_Rating), 2) AS avg_rating
FROM customers
GROUP BY Age_Group
ORDER BY Age_Group;
```

Result:

Age_Group	customers	avg_spend	avg_prev_purchases	avg_cvs	promo_pct	subscription_pct	avg_rating
18-24	486	60.20	24.2	0.415	42.6	25.7	3.81
25-34	755	60.13	25.0	0.426	43.2	26.6	3.76
35-44	729	59.62	25.5	0.431	43.3	26.7	3.75
45-54	752	60.33	24.9	0.43	41.5	29.1	3.71
55-64	751	58.72	26.1	0.426	43.8	27.6	3.72
65+	427	59.70	26.4	0.435	43.8	24.8	3.78

Key Findings

- The **65+ age group** has the highest Customer Value Score (**0.435**) and highest average previous purchases (**26.4**), indicating strong loyalty.
- The 45–54 age group shows the highest average spend (\$60.33); the 35–44 group shows strong customer value (0.431).
- Promo dependency is very similar across all age groups (**41–44%**), suggesting age is not a major driver of discount reliance.

Conclusion

Customer value generally increases with age, with 35–44 and 65+ customers representing the strongest segments for retention and long-term value growth.

Analysis 2: Gender Analysis

Objective:

Evaluate differences in spending, customer value, purchase behavior, and promo dependency between genders.

SQL Query:

```
USE customer_retention;
SELECT
    Gender,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS promo_pct,
    ROUND(AVG(Customer_Value_Score), 3) AS avg_cvs,
    ROUND(AVG(Subscription_Status) * 100, 1) AS subscription_pct
FROM customers
GROUP BY Gender;
```

Result:

Gender	customers	avg_spend	avg_prev_purchases	promo_pct	avg_cvs	subscription_pct
Male	2652	59.54	25.7	63.2	0.429	39.7
Female	1248	60.25	24.6	0.0	0.423	0.0

Key Findings

- Female customers spend slightly more (\$60.25 vs \$59.54).
- Male customers have slightly higher customer value (0.429 vs 0.423) and repeat purchases.
- Promo dependency is 0% for females and 63.2% for males — a complete split, not a gradual difference (data artifact, not a behavioral finding).

Conclusion: Customer value is similar across genders. The promo-dependency split is a data artifact (see note above) — segments built on Promo_Dependent (Champion, Promo Convert Target, etc.) inherit this pattern.

Analysis 3: Age Group × Category Analysis

Objective:

Identify category preferences and purchasing behavior across different age groups.

SQL Query:

```
SELECT
    Age_Group,
    Category,
    COUNT(*)                                AS customers,
    ROUND(AVG(Purchase_Amount), 2)         AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1)     AS avg_prev_purchases
FROM customers
GROUP BY Age_Group, Category
ORDER BY Age_Group, customers DESC;
```

Result:

Age_Group	Category	customers	avg_spend	avg_prev_purchases
18-24	Clothing	236	59.53	23.1
18-24	Accessories	142	60.92	25.4
18-24	Footwear	61	60.70	24.3
18-24	Outerwear	47	60.74	26.4
25-34	Clothing	340	60.20	25.4
25-34	Accessories	248	59.78	25.3
25-34	Footwear	115	63.50	23.4
25-34	Outerwear	52	53.90	25.0
35-44	Clothing	314	59.33	25.7
35-44	Accessories	249	60.26	25.0
35-44	Footwear	109	58.61	26.2
35-44	Outerwear	57	60.33	24.8
45-54	Clothing	325	60.34	25.0
45-54	Accessories	221	60.90	25.1
45-54	Footwear	139	60.16	24.9
45-54	Outerwear	67	58.78	24.0
55-64	Clothing	327	59.80	25.9
55-64	Accessories	236	58.52	26.4
55-64	Footwear	123	58.11	25.7
55-64	Outerwear	65	55.11	26.6
65+	Clothing	195	61.27	25.7
65+	Accessories	144	58.68	27.9
65+	Footwear	52	61.33	28.0
65+	Outerwear	36	52.97	22.0

Key Findings

- **Clothing** is the most popular category across all age groups, indicating broad customer appeal.
- Customers aged **55+** generally show higher previous purchases, suggesting stronger repeat-buying behavior.
- Average spending is fairly consistent across age groups and categories, mostly ranging between **\$58–\$63**.

Conclusion

Clothing acts as the primary acquisition category across all demographics, while older customers (55+) demonstrate stronger loyalty through higher repeat purchases, making them valuable retention targets.

SECTION 7: Promotion Dependency Deep Dive

Business Question

Who relies on discounts and who can be transitioned away from promotions?

Analysis 1: Promo Dependency by Subscription Status

Objective:

Examine whether subscribers are more or less dependent on promotions.

SQL Query:

```
SELECT
    Subscription_Status,
    Promo_Dependent,
    COUNT(*)                                AS customers,
    ROUND(AVG(Purchase_Amount), 2)          AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1)      AS avg_prev_purchases,
    ROUND(AVG(Customer_Value_Score), 3)     AS avg_cvs
FROM customers
GROUP BY Subscription_Status, Promo_Dependent
ORDER BY Subscription_Status DESC, Promo_Dependent DESC;
```

Result:

Subscription_Status	Promo_Dependent	customers	avg_spend	avg_prev_purchases	avg_cvs
1	1	1053	59.49	26.1	0.436
0	1	624	58.92	25.2	0.419
0	0	2223	60.13	25.1	0.425

Key Findings

- Subscribers have slightly higher customer value (**0.436**) and purchase history (**26.1 purchases**) than non-subscribers.
- Non-promo customers generate the highest average spend (**\$60.13**).
- Both subscriber and non-subscriber groups contain promo-dependent customers.

Conclusion

Subscriptions are associated with slightly stronger loyalty and customer value, while non-promo customers remain the most valuable segment for sustainable revenue.

Analysis 2: Promo Dependency by Purchase Frequency

Objective:

Understand how discount dependency changes across purchase frequency levels.

SQL Query:

```
SELECT
    Frequency_of_Purchases,
    Frequency_Score,
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS promo_pct,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend
FROM customers
GROUP BY Frequency_of_Purchases, Frequency_Score
ORDER BY Frequency_Score DESC;
```


Result:

Frequency_of_Purchases	Frequency_Score	promo_pct	customers	avg_spend
Weekly	52	43.4	539	58.97
Fortnightly	26	43.1	1089	59.88
Monthly	12	42.9	553	59.33
Quarterly	4	43.5	1147	60.03
Annually	1	41.6	572	60.17

Key Findings

- Promo dependency is flat across frequency levels (41.6%–43.5%), and the highest rate is Quarterly, not Weekly — no clear trend by frequency.

Conclusion: Purchase frequency does not meaningfully predict promo dependency in this dataset.

Analysis 3: Promotional Sunset Plan Candidates

Objective:

Profile the "Promo Convert Target" segment and assess its suitability for a promo-sunset strategy.

SQL Query:

```
SELECT
    Customer_Segment,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Frequency_Score), 1) AS avg_frequency,
    ROUND(AVG(Customer_Value_Score), 3) AS avg_cvs,
    ROUND(AVG(Review_Rating), 2) AS avg_rating
FROM customers
WHERE Customer_Segment = 'Promo Convert Target'
GROUP BY Customer_Segment;
```

Result:

Customer_Segment	customers	avg_spend	avg_prev_purchases	avg_frequency	avg_cvs	avg_rating
Promo Convert Target	564	65.59	39.5	25.3	0.619	3.75

Key Findings

- The Promo Convert Target segment contains 564 customers (14.5%).
- They have high average spend (\$65.59) and the highest customer value score (0.619).
- They also show strong purchase history (39.5 previous purchases) and frequent buying behavior.

Conclusion

The Promo Convert Target segment is the best candidate for a promo-sunset strategy, as these customers are already highly engaged and valuable, making them likely to continue purchasing with reduced discounts.

Analysis 4: Detailed Profile of Sunset Candidates

Objective:

Understand the demographic and behavioral characteristics of the highest-value promo-convert customers.

SQL Query:

```
SELECT
    Age,
    Gender,
    Location,
    Category,
    Purchase_Amount,
    Previous_Purchases,
    Frequency_of_Purchases,
    Frequency_Score,
    Loyalty_Label_A,
    Value_Tier,
    Customer_Value_Score,
    Tenure_Band,
    Review_Rating,
    Satisfaction_Flag
FROM customers
WHERE Customer_Segment = 'Promo Convert Target'
ORDER BY Customer_Value_Score DESC
LIMIT 50;
```

Result:

Age	Gender	Location	Category	Purchase_Amo	Previous_Purcha	Frequency_of_Purcha	Frequen	Loyalty_Label	Value_Ti	Customer_Value_Score	Tenure_I_Review	Satisfaction_Flag
46	Male	Louisiana	Outerwear	39	50	Weekly	52	Loyal	Platinum	0.9375	Veteran	4.6 High
21	Male	Rhode Island	Footwear	30	49	Weekly	52	Loyal	Platinum	0.9668	Veteran	3.5 Medium
34	Male	Georgia	Clothing	31	48	Weekly	52	Loyal	Platinum	0.9612	Veteran	3.6 Medium
55	Male	WYoming	Clothing	94	47	Weekly	52	Loyal	Platinum	0.9605	Veteran	2.5 Low
59	Male	Arizona	Accessori	84	50	Weekly	52	Loyal	Platinum	0.96	Veteran	2.8 Low
64	Male	Illinois	Accessori	95	45	Weekly	52	Loyal	Platinum	0.9467	Veteran	2.6 Low
29	Male	Illinois	Clothing	91	46	Weekly	52	Loyal	Platinum	0.9448	Veteran	3.4 Medium
57	Male	Louisiana	Clothing	89	46	Weekly	52	Loyal	Platinum	0.9398	Veteran	3.1 Medium
50	Male	North Carolin	Accessori	75	50	Weekly	52	Loyal	Platinum	0.9375	Veteran	3.3 Medium
35	Male	Washington	Accessori	97	43	Weekly	52	Loyal	Platinum	0.9354	Veteran	3.3 Medium
55	Male	Maryland	Accessori	80	47	Weekly	52	Loyal	Platinum	0.9255	Veteran	4.9 High
45	Male	Oregon	Accessori	95	41	Weekly	52	Loyal	Platinum	0.914	Veteran	4 High
42	Male	Ohio	Footwear	72	48	Weekly	52	Loyal	Platinum	0.9137	Veteran	2.7 Low
37	Male	Rhode Island	Clothing	97	40	Weekly	52	Loyal	Platinum	0.9109	Veteran	3.7 Medium
30	Male	New Mexico	Footwear	60	50	Weekly	52	Loyal	Platinum	0.9	Veteran	2.8 Low
41	Male	Mississippi	Accessori	73	46	Weekly	52	Loyal	Platinum	0.8998	Veteran	4.7 High
29	Male	Washington	Accessori	59	50	Weekly	52	Loyal	Platinum	0.8975	Veteran	2.7 Low
50	Male	Georgia	Clothing	70	46	Weekly	52	Loyal	Platinum	0.8923	Veteran	4.9 High
20	Male	Oregon	Footwear	60	49	Weekly	52	Loyal	Platinum	0.8918	Veteran	3.3 Medium
32	Male	Alaska	Accessori	66	47	Weekly	52	Loyal	Platinum	0.8905	Veteran	4.1 High
33	Male	Arizona	Accessori	79	43	Weekly	52	Loyal	Platinum	0.8904	Veteran	2.7 Low
31	Male	Oklahoma	Clothing	88	40	Weekly	52	Loyal	Platinum	0.8884	Veteran	4.4 High
47	Male	Oregon	Clothing	83	41	Weekly	52	Loyal	Platinum	0.884	Veteran	2.7 Low
42	Male	North Dakota	Clothing	56	49	Weekly	52	Loyal	Platinum	0.8818	Veteran	3.7 Medium
25	Male	Alaska	Clothing	68	45	Weekly	52	Loyal	Platinum	0.8792	Veteran	4.8 High
68	Male	Louisiana	Accessori	93	37	Weekly	52	Loyal	Platinum	0.8764	Establish	4.3 High
58	Male	Pennsylvania	Accessori	76	42	Weekly	52	Loyal	Platinum	0.8747	Veteran	3 Medium
37	Male	Florida	Clothing	62	46	Weekly	52	Loyal	Platinum	0.8723	Veteran	4.5 High
54	Male	Massachusetts	Clothing	94	36	Weekly	52	Loyal	Platinum	0.8707	Establish	3.6 Medium
36	Male	Ohio	Footwear	60	46	Weekly	52	Loyal	Platinum	0.8673	Veteran	3.9 Medium
23	Male	Tennessee	Accessori	89	37	Weekly	52	Loyal	Platinum	0.8664	Establish	4.9 High
49	Male	WYoming	Outerwear	48	49	Weekly	52	Loyal	Platinum	0.8618	Veteran	4 High
47	Male	Pennsylvania	Accessori	100	33	Weekly	52	Loyal	Platinum	0.8612	Establish	4.8 High
28	Male	Wisconsin	Accessori	67	43	Weekly	52	Loyal	Platinum	0.8604	Veteran	4.1 High
70	Male	Minnesota	Clothing	60	45	Weekly	52	Loyal	Platinum	0.8592	Veteran	3.2 Medium
42	Male	Maryland	Outerwear	82	38	Weekly	52	Loyal	Platinum	0.857	Establish	4 High
38	Male	Indiana	Footwear	68	42	Weekly	52	Loyal	Platinum	0.8547	Veteran	4.5 High
39	Male	Indiana	Clothing	93	34	Weekly	52	Loyal	Platinum	0.8519	Establish	3.9 Medium
62	Male	West Virginia	Accessori	47	48	Weekly	52	Loyal	Platinum	0.8512	Veteran	2.8 Low
50	Male	Texas	Accessori	73	40	Weekly	52	Loyal	Platinum	0.8509	Veteran	4.9 High
67	Male	Hawaii	Clothing	91	34	Weekly	52	Loyal	Platinum	0.8469	Establish	4.1 High
37	Male	Arkansas	Accessori	81	37	Weekly	52	Loyal	Platinum	0.8464	Establish	4.3 High
21	Male	North Carolin	Accessori	84	36	Weekly	52	Loyal	Platinum	0.8457	Establish	4.9 High
39	Male	Kentucky	Accessori	53	45	Weekly	52	Loyal	Platinum	0.8417	Veteran	4.6 High
62	Male	New Jersey	Accessori	41	48	Weekly	52	Loyal	Platinum	0.8362	Veteran	3.2 Medium
54	Male	Kentucky	Outerwear	76	37	Weekly	52	Loyal	Platinum	0.8339	Establish	3.6 Medium
58	Male	Colorado	Clothing	33	50	Weekly	52	Loyal	Platinum	0.8325	Veteran	3.7 Medium
30	Male	New Mexico	Accessori	58	42	Weekly	52	Loyal	Platinum	0.8297	Veteran	4.7 High
56	Male	Virginia	Accessori	46	45	Weekly	52	Loyal	Platinum	0.8242	Veteran	4.9 High
34	Male	New Jersey	Footwear	45	45	Weekly	52	Loyal	Platinum	0.8217	Veteran	4.2 High

Key Findings

- Top sunset candidates are predominantly Loyal, Platinum-tier customers with very high Customer Value Scores (0.80–1.00).
- Most have 40–50 previous purchases and weekly purchase frequency, indicating strong purchasing habits.
- Customer ratings are generally positive (3.0–5.0) and most belong to Established/Veteran tenure bands.

Conclusion

These customers already demonstrate strong loyalty, high value, and consistent purchase behavior. They are ideal candidates for a gradual discount reduction strategy, as their purchasing decisions appear driven more by habit and brand loyalty than by promotions.

SECTION 8: Customer Segment Analysis

Business Question

How do customer segments differ in value, spending, and behavior?

Analysis 1: Complete Segment Summary

Objective:

Compare all customer segments based on spending, loyalty, customer value, promo dependency, and engagement.

SQL Query:

```
SELECT
    Customer_Segment,
    COUNT(*) AS customer_count,
    ROUND(COUNT(*) * 100.0 /
          (SELECT COUNT(*) FROM customers), 1) AS pct_of_base,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Frequency_Score), 1) AS avg_frequency,
    ROUND(AVG(Customer_Value_Score), 3) AS avg_cvs,
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS promo_pct,
    ROUND(AVG(Review_Rating), 2) AS avg_rating,
    ROUND(AVG(Subscription_Status) * 100, 1) AS subscription_pct
FROM customers
GROUP BY Customer_Segment
ORDER BY avg_cvs DESC;
```

Result:

Customer_Segment	customer_count	pct_of_base	avg_spend	avg_prev_purchases	avg_frequency	avg_cvs	promo_pct	avg_rating	subscription_pct
Promo Convert Target	564	14.5	65.59	39.5	25.3	0.619	100.0	3.75	66.0
Champion	873	22.4	65.64	34.3	29.5	0.61	0.0	3.77	0.0
Loyal Customer	145	3.7	37.81	39.6	7.5	0.411	0.0	3.76	0.0
Low Engagement	862	22.1	57.84	25.7	9.8	0.365	44.4	3.83	28.2
At Risk	199	5.1	54.80	19.8	11.6	0.324	100.0	2.72	65.8
Growth Potential	726	18.6	57.73	11.6	14.4	0.286	0.0	3.73	0.0
Discount Dependent	531	13.6	57.69	12.0	10.8	0.261	100.0	4.01	57.8

Key Findings

- Promo Convert Target has the highest customer value (CVS 0.619) and spend (\$65.59) but is fully promo-dependent (100%).
- Champion customers represent the largest high-value segment (22.4% of customers) with strong spend (\$65.64) and zero promo dependency.
- At Risk customers have the lowest rating (2.72) and low customer value (0.324).
- Loyal Customer has the lowest avg. spend of any segment (\$37.81) — the label is based on frequency/history only, not spend.

Conclusion

Champions are the most valuable organic customers, while Promo Convert Targets are the best candidates for a promo-sunset strategy due to their high value but complete reliance on discounts.

Analysis 2: Segment × Category Analysis

Objective:

Identify category preferences across customer segments.

SQL Query:

```
SELECT
    Customer_Segment,
    Category,
    COUNT(*)                                AS customers,
    ROUND(AVG(Purchase_Amount), 2)         AS avg_spend
FROM customers
GROUP BY Customer_Segment, Category
ORDER BY Customer_Segment, customers DESC;
```

Result:

Customer_Segment	Category	customers	avg_spend
At Risk	Outerwear	17	46.76
Champion	Clothing	396	65.66
Champion	Accessories	270	65.88
Champion	Footwear	138	65.33
Champion	Outerwear	69	65.26
Discount Dependent	Clothing	234	57.96
Discount Dependent	Accessories	176	55.65
Discount Dependent	Footwear	82	62.74
Discount Dependent	Outerwear	39	54.64
Growth Potential	Clothing	321	56.32
Growth Potential	Accessories	228	60.12
Growth Potential	Footwear	113	56.57
Growth Potential	Outerwear	64	58.30
Low Engagement	Clothing	380	59.37
Low Engagement	Accessories	275	57.85
Low Engagement	Footwear	130	57.33

Low Engagement	Outerwear	77	51.06
Loyal Customer	Clothing	66	39.27
Loyal Customer	Accessories	50	36.28
Loyal Customer	Footwear	18	35.89
Loyal Customer	Outerwear	11	39.18
Promo Convert Tar...	Clothing	250	65.73
Promo Convert Tar...	Accessories	179	64.84
Promo Convert Tar...	Footwear	88	67.63
Promo Convert Tar...	Outerwear	47	63.85

Key Findings

- Champion segment shows the highest spending across all categories (~\$65+), especially in Clothing and Accessories.
- Promo Convert Target also has high spend (~\$63–\$67), confirming strong value but discount-driven behavior.
- Low Engagement and Loyal segments show comparatively lower spend (~\$35–\$51), indicating weaker monetization.

Conclusion

High-value segments (Champion + Promo Convert Target) dominate category spending, making them the core revenue drivers, while Loyal and Low Engagement groups represent optimization opportunities.

Analysis 3: Segment × Gender Analysis

Objective:

Examine gender distribution and customer value across segments.

SQL Query:

```
SELECT
    Customer_Segment,
    Gender,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Customer_Value_Score), 3) AS avg_cvs
FROM customers
GROUP BY Customer_Segment, Gender
ORDER BY Customer_Segment, customers DESC;
```

Result:

Customer_Segment	Gender	customers	avg_spend	avg_cvs
At Risk	Male	199	54.80	0.324
Champion	Female	488	65.14	0.603
Champion	Male	385	66.28	0.618
Discount Dependent	Male	531	57.69	0.261
Growth Potential	Female	416	58.13	0.287
Growth Potential	Male	310	57.18	0.284
Low Engagement	Male	598	56.53	0.391
Low Engagement	Female	264	60.80	0.307
Loyal Customer	Female	80	39.63	0.412
Loyal Customer	Male	65	35.58	0.41
Promo Convert Tar...	Male	564	65.59	0.619

Key Findings

- Both male and female Champion customers have the highest customer value (CVS ~0.60+) and spending (~\$65–\$6).
- Promo Convert Target is entirely male in this segment and shows the highest customer value (CVS 0.619).
- Customer value differences within segments are small, indicating behavior matters more than gender.

Conclusion

Customer segment is a stronger driver of value than gender. High-value customers are concentrated in the Champion and Promo Convert Target segments regardless of gender

SECTION 9: Ideal Customer Profile

Business Question

What characteristics define the highest-value organic customers?

Analysis 1: Ideal Customer Demographics

Objective:

Profile high-value customers who purchase without relying on promotions.

SQL Query:

```
SELECT
    Gender,
    ROUND(AVG(Age), 0) AS avg_age,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Frequency_Score), 1) AS avg_frequency,
    ROUND(AVG(Review_Rating), 2) AS avg_rating,
    ROUND(AVG(Subscription_Status) * 100, 1) AS subscription_pct,
    COUNT(*) AS customer_count
FROM customers
WHERE High_Value_Flag = 1
    AND Promo_Dependent = 0
GROUP BY Gender
ORDER BY avg_spend DESC;
```

Result:

Gender	avg_age	avg_spend	avg_prev_purchases	avg_frequency	avg_rating	subscription_pct	customer_count
Male	45	67.70	34.4	28.8	3.8	0.0	408
Female	44	66.71	33.2	28.4	3.74	0.0	522

Key Findings

- High-value organic customers are predominantly aged 44–45 years.
- Both genders show very similar behavior, with average spend around \$67 and 33–34 previous purchases.
- These customers purchase frequently (28+ frequency score) and maintain good ratings (3.7–3.8).

Conclusion

The ideal customer is a mid-aged, high-frequency buyer with a long purchase history who purchases without relying on discounts, regardless of gender.

Analysis 2: Ideal Customer Top Category

Objective:

Identify the most common product categories among ideal customers.

SQL Query:

```
SELECT
    Category,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend
FROM customers
WHERE High_Value_Flag = 1
    AND Promo_Dependent = 0
GROUP BY Category
ORDER BY customers DESC;
```

Result:

Category	customers	avg_spend
Clothing	429	67.53
Accessories	285	67.04
Footwear	145	66.66
Outerwear	71	66.18

Key Findings

- Clothing is the most common category among ideal customers (429 customers).
- Clothing and Accessories account for the majority of high-value organic customers.
- Average spend is consistently high across all categories (\$66–68).

Conclusion

Clothing is the primary category of ideal customers, making it the strongest category for acquiring and retaining high-value, non-promo-dependent buyers.

Analysis 3: Ideal Customer Top States

Objective:

Identify geographic concentrations of ideal customers.

SQL Query:

```
SELECT
    Location,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Customer_Value_Score), 3) AS avg_cvs
FROM customers
WHERE High_Value_Flag = 1
    AND Promo_Dependent = 0
GROUP BY Location
ORDER BY customers DESC
LIMIT 10;
```

Result:

Location	customers	avg_spend	avg_cvs
Alabama	28	65.36	0.637
Georgia	27	63.70	0.588
Minnesota	25	68.28	0.628
Tennessee	25	70.56	0.649
Alaska	25	72.40	0.605
Pennsylvania	24	73.08	0.622
New York	23	66.48	0.658
Nevada	23	74.22	0.619
Arizona	23	71.74	0.608
Maryland	23	64.43	0.591

Key Findings

- States such as Nevada, Pennsylvania, Alaska, and Tennessee have the highest-spending ideal customers (\$70+ average spend).
- These states also show strong customer value scores (CVS ~0.60–0.66).
- Ideal customers are distributed across multiple states rather than concentrated in a single region.

Conclusion

States like **Nevada, Tennessee, Alaska, and Pennsylvania** represent attractive markets for targeting and retaining high-value, non-promo-dependent customers.

Analysis 4: Ideal Customer Payment Method

Objective:

Determine preferred payment methods of ideal customers.

SQL Query:

```
SELECT
    Payment_Method,
    COUNT(*)                                AS customers,
    ROUND(AVG(Purchase_Amount), 2)         AS avg_spend
FROM customers
WHERE High_Value_Flag = 1
    AND Promo_Dependent = 0
GROUP BY Payment_Method
ORDER BY customers DESC;
```

Result:

Payment_Method	customers	avg_spend
Cash	170	68.38
Credit Card	167	66.65
PayPal	166	65.42
Debit Card	147	67.57
Venmo	141	67.28
Bank Transfer	139	67.69

Key Findings

- Cash is the most preferred payment method among ideal customers (170 customers) with the highest average spend (\$68.38).
- Digital methods like Credit Card, Debit Card, PayPal, and Bank Transfer show very similar spend levels (\$65–68).
- No major variation in spending across payment methods, indicating payment type is not a strong value driver.

Conclusion : Ideal high-value customers use multiple payment methods evenly, with a slight preference for cash and card-based payments, and payment method does not significantly impact customer value.

Analysis 5: Ideal Customer Season Preference

Objective:

Identify seasons generating the highest engagement among ideal customers.

SQL Query:

```
SELECT
    Season,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend
FROM customers
WHERE High_Value_Flag = 1
    AND Promo_Dependent = 0
GROUP BY Season
ORDER BY customers DESC;
```

Result:

Season	customers	avg_spend
Winter	249	67.81
Spring	234	65.18
Summer	224	67.13
Fall	223	68.48

Key Findings

- **Winter** has the highest number of ideal customers (**249 customers**).
- Average spend remains consistently high across all seasons (**\$65–68**).
- Customer distribution is fairly balanced, with no single season dominating.

Conclusion

Ideal customers are active throughout the year, with a slight concentration in **Winter**, making it the strongest season for engaging high-value, non-promo-dependent customers.

Analysis 6: At-Risk Segment Profile

Objective:

Understand the behavior and value of customers classified as At Risk.

SQL Query:

```
SELECT
    COUNT(*) AS at_risk_customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Review_Rating), 2) AS avg_rating,
    ROUND(AVG(Frequency_Score), 1) AS avg_frequency
FROM customers
WHERE Customer_Segment = 'At Risk';
```

Result:

at_risk_customers	avg_spend	avg_prev_purchases	avg_rating	avg_frequency
199	54.80	19.8	2.72	11.6

Key Findings

- The At Risk segment contains 199 customers.
- Average spend is relatively low at \$54.80.
- Customers have only 19.8 previous purchases and a low frequency score of 11.6.
- Average rating is the lowest seen so far (2.72), indicating poor satisfaction.

Conclusion

At-Risk customers show low engagement, low satisfaction, and declining purchase activity. They should be prioritized for retention campaigns before they churn completely.

SECTION 10: Champion Segment Deep Dive

Business Question

Who are the brand's best customers and what makes them valuable?

Analysis 1: Champion Profile

Objective:

Profile Champion customers by demographics and purchasing behavior.

SQL Query:

```
SELECT
    Gender,
    ROUND(AVG(Age), 0) AS avg_age,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Frequency_Score), 1) AS avg_frequency,
    ROUND(AVG(Review_Rating), 2) AS avg_rating,
    ROUND(AVG(Subscription_Status) * 100, 1) AS subscription_pct,
    COUNT(*) AS customers
FROM customers
WHERE Customer_Segment = 'Champion'
GROUP BY Gender;
```

Result:

Gender	avg_age	avg_spend	avg_prev_purchases	avg_frequency	avg_rating	subscription_pct	customers
Male	44	66.28	35.0	29.7	3.79	0.0	385
Female	44	65.14	33.8	29.4	3.75	0.0	488

Key Findings

- Champion customers are evenly distributed across genders (488 females, 385 males).
- Both groups have very high spending (~\$65–66) and strong purchase history (34+ previous purchases).
- Frequency scores are the highest observed (~29.5), indicating very active customers.
- Customer behavior is nearly identical across genders.

Conclusion

Champion customers represent the most valuable and loyal segment regardless of gender. Retention efforts should focus on maintaining engagement and rewarding their high purchase frequency.

Analysis 2: Champion Top Locations

Objective:

Identify where Champion customers are concentrated.

SQL Query:

```
SELECT
    Location,
    COUNT(*) AS champion_customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend
FROM customers
WHERE Customer_Segment = 'Champion'
GROUP BY Location
ORDER BY champion_customers DESC
LIMIT 10;
```

Result:

Location	champion_customers	avg_spend
Alabama	28	65.36
Georgia	27	63.70
Minnesota	24	67.33
Alaska	24	71.38
Arizona	23	71.74
Tennessee	23	68.96
Pennsylvania	22	71.59
New York	22	65.27
Nevada	22	73.86
Maryland	21	61.29

Key Findings

- Champion customers are spread across multiple states, with Alabama (28) and Georgia (27) having the highest counts.
- States such as Nevada (\$73.86), Pennsylvania (\$71.59), and Arizona (\$71.74) show the highest average spend among Champions.
- Champion customers are not concentrated in a single region, indicating broad geographic strength.

Conclusion

Champion customers are distributed across many states, but Nevada, Pennsylvania, and Arizona stand out as high-value markets. These locations should be prioritized for loyalty and retention initiatives.

Analysis 3: Champion Top Categories

Objective:

Identify product categories most popular among Champion customers.

SQL Query:

```
SELECT
    Category,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases
FROM customers
WHERE Customer_Segment = 'Champion'
GROUP BY Category
ORDER BY customers DESC;
```

Result:

Category	customers	avg_spend	avg_prev_purchases
Clothing	396	65.66	33.6
Accessories	270	65.88	35.0
Footwear	138	65.33	34.3
Outerwear	69	65.26	35.6

Key Findings

- Clothing is the most popular category among Champion customers (396 customers).
- Accessories is the second-largest category (270 customers).
- Average spend is consistently high across all categories (~\$65–66).
- Champion customers have strong purchase history in every category (33–36 previous purchases).

Conclusion

Champion customers are most concentrated in **Clothing and Accessories**, making these categories ideal for loyalty programs, premium offerings, and personalized retention campaigns.

SECTION 11: Executive Summary

Business Question

What are the key business insights and strategic recommendations?

Analysis 1: Overall Brand Health

Objective:

Measure overall customer value, revenue, loyalty, and promo dependency.

SQL Query:

```
SELECT
    COUNT(*) AS total_customers,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(SUM(Purchase_Amount), 0) AS total_revenue,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases,
    ROUND(AVG(Review_Rating), 2) AS avg_rating,
    ROUND(AVG(Subscription_Status) * 100, 1) AS subscription_pct,
    ROUND(SUM(Promo_Dependent) * 100.0 / COUNT(*), 1) AS overall_promo_pct,
    ROUND(SUM(High_Value_Flag) * 100.0 / COUNT(*), 1) AS high_value_pct
FROM customers;
```

Result:

total_customers	avg_spend	total_revenue	avg_prev_purchases	avg_rating	subscription_pct	overall_promo_pct	high_value_pct
3900	59.76	233081	25.4	3.75	27.0	43.0	42.4

Key Findings

- The business has 3,900 customers generating \$233,081 total revenue.
- Average customer spend is \$59.76 with 25.4 previous purchases, indicating healthy repeat buying.
- 42.4% of customers are high-value customers.
- 43.0% of customers are promo-dependent, while the majority (57%) purchase without discounts.

Conclusion

The brand demonstrates a strong customer base with a large share of high-value and organically purchasing customers. The business is well-positioned to reduce promotional spending through targeted retention and promo-sunset strategies while maintaining revenue growth.

Analysis 2: Champion vs Rest of Base

Objective:

Compare the contribution of Champion customers against the rest of the customer base.

SQL Query:

```
SELECT
    CASE WHEN Customer_Segment = 'Champion'
        THEN 'Champion'
        ELSE 'Rest of Base'
    END AS customer_group,
    COUNT(*) AS customers,
    ROUND(SUM(Purchase_Amount) * 100.0 /
        (SELECT SUM(Purchase_Amount)
         FROM customers), 1) AS revenue_share_pct,
    ROUND(AVG(Purchase_Amount), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_prev_purchases
FROM customers
GROUP BY customer_group;
```

Result:

customer_group	customers	revenue_share_pct	avg_spend	avg_prev_purchases
Rest of Base	3027	75.4	58.07	22.8
Champion	873	24.6	65.64	34.3

Key Findings

- Champions represent only 22.4% of customers (873 customers) but generate 24.6% of total revenue.
- Champion customers spend significantly more (\$65.64 vs \$58.07) than the rest of the customer base.
- Champions have much stronger purchase history (34.3 vs 22.8 previous purchases).

Conclusion

Champion customers are a relatively small segment but contribute disproportionately to revenue. Retaining and expanding this segment should be a top business priority, as they drive higher spending and repeat purchases.

Analysis 3: Promo Dependency Revenue Impact

Objective:

Estimate the financial impact of discount-driven purchases.

SQL Query:

```
SELECT
    ROUND(SUM(CASE WHEN Promo_Dependent = 1
        THEN Purchase_Amount ELSE 0 END), 0) AS promo_revenue,
    ROUND(SUM(CASE WHEN Promo_Dependent = 0
        THEN Purchase_Amount ELSE 0 END), 0) AS organic_revenue,
    ROUND(SUM(CASE WHEN Promo_Dependent = 1
        THEN Purchase_Amount * 0.15 ELSE 0 END),0)AS estimated_discount_cost,
    ROUND(SUM(Purchase_Amount), 0) AS total_revenue
FROM customers;
```

Result:

promo_revenue	organic_revenue	estimated_discount_cost	total_revenue
99411	133670	14912	233081

Key Findings

- Organic revenue (\$133,670) is significantly higher than promo-driven revenue (\$99,411).
- Estimated discount cost is \$14,912, representing potential savings if promo dependency is reduced.
- More than half of total revenue comes from customers who purchase without discounts.

Conclusion

The business relies more on organic purchasing than promotions. Reducing discount usage among selected customer segments could lower costs while maintaining a large portion of revenue.

